TECHNICAL ARCHITECTURE & IMPLEMENTATION BLUEPRINT

TransportLah — System Architecture & Delivery Plan

A gamified, privacy-first green commute rewards platform for Singapore

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1. Executive Summary & Market Differentiator

Singapore has mature transport utilities (Citymapper, MyTransport.SG, SG BusLeh, and SimplyGo). TransportLah does not seek to duplicate basic journey navigation. Instead, it leverages real-time LTA DataMall transit feeds to establish Singapore's first **verified green mobility reward and challenge platform**:

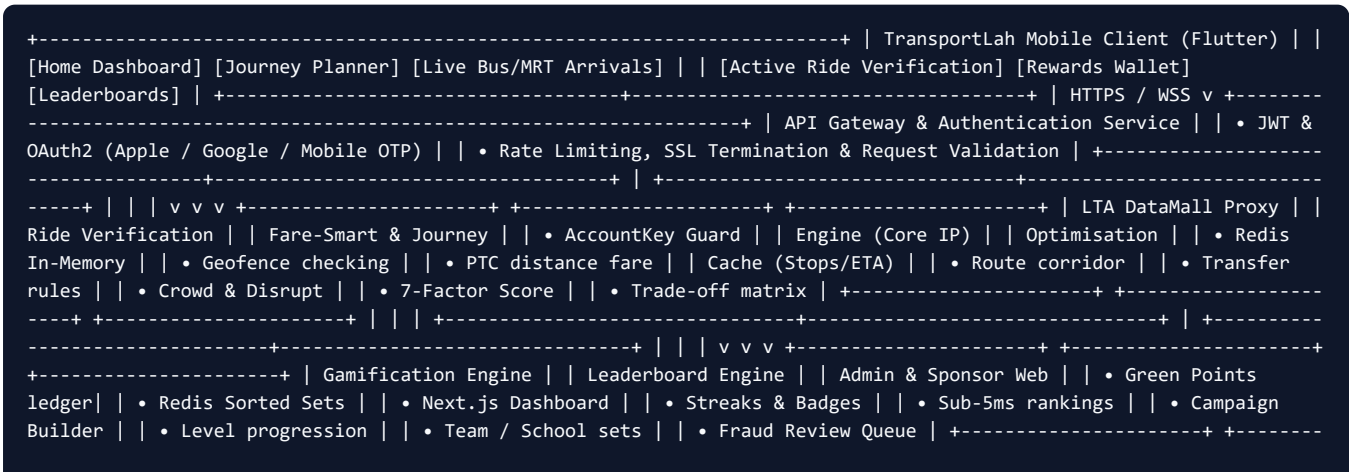
The Core Product Loop:
Plan Ride → Verify Ride (Privacy-First) → Earn Green Points → Join Sponsor Contests → Climb Leaderboard → Redeem Impact Rewards

2. Cloud-Native Build & Hardware Architecture (Zero Local Overhead)

To ensure absolute reliability, rapid development velocity, and complete independence from local development hardware:

- **Cross-Platform Mobile Core (Flutter / Dart):** A single unified codebase delivering native 60fps performance across both iOS and Android. Zero duplication of business logic or UI states.
- **Cloud macOS CI/CD for iOS (GitHub Actions):** Apple requires macOS/Xcode for iOS compilation. We utilize cloud-hosted macOS virtual runners on GitHub Actions to compile `.ipa` builds and deploy automatically to **Apple TestFlight**.
- **Cloud Linux CI/CD for Android:** Automated builds compile and sign release `.aab` bundles for direct release to Google Play Console.
- **Cloud Containerized Backend:** Microservices containerized in Docker, running on scalable AWS / GCP managed infrastructure (ECS Fargate + RDS PostgreSQL/PostGIS + ElastiCache Redis).

3. End-to-End System Architecture



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-+ | Data Layer | | • PostgreSQL 16 + PostGIS: Spatial stop coordinates, routes, users | | • Redis 7: Active ride
ephemeral states, fast caches, leaderboards | | • AWS S3: Campaign banners, sponsor creatives, report exports | +-----
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4. Lightweight Ride Verification Engine (The Core IP)

Unlike battery-draining navigation apps that constantly track users, TransportLah implements a **privacy-first confidence scoring algorithm**. Location sampling occurs only during active rides, and raw traces are permanently purged within 48–72 hours.

Multi-Factor Confidence Scoring Matrix (Total: 100 Points)

Signal	Verification Metric & Logic	Weight
1. Stop Proximity	Boarding coordinates fall within 50m of an official LTA bus stop or MRT station (PostGIS `ST_DWithin`).	15 pts
2. Service Match	User's selected bus service officially serves the boarding stop.	10 pts
3. Live Arrival Correlation	Cross-referenced with LTA live arrival API; confirms vehicle arrived within ±3 min of check-in.	20 pts
4. Route Corridor Following	User trajectory tracks official stop sequence (Stop N → N+1 → N+2).	25 pts
5. Kinematic Plausibility	Speed profile matches public transit dynamics (12–65 km/h; rejects walking or private car speeds).	10 pts
6. Corridor Tolerance	User coordinates remain within a 150m buffer of the route line geometry.	10 pts
7. Alighting Verification	Ride ends in proximity to a downstream official stop along the same line.	10 pts

Verification Tiers & Point Allocations

Confidence Score	Verification Status	Points & Contest Eligibility
80 – 100	**Verified Ride**	100% Green Points awarded + Fully eligible for public/sponsor contests.
60 – 79	**Probable Ride**	Partial points awarded (50%); personal streak preserved; not contest eligible.
40 – 59	Low Confidence	Logged in personal commuter statistics only; 0 Green Points.
< 40	**Rejected / Flagged**	Zero points; repeated triggers routed to Admin Fraud Review queue.

5. Database & Spatial Architecture (PostGIS + Redis)

Key Spatial & Ephemeral Models:

- `stops` (id, code, name, geom GEOMETRY(Point, 4326), mode) — Indexed with `GIST(geom)` for sub-millisecond geofence queries.
- `route_corridors` (service_no, direction, path GEOMETRY(LineString, 4326)) — Route corridor matching and buffer validation.
- `rides` (id, user_id, mode, service_no, start_stop, end_stop, start_time, end_time, confidence_score, status).
- `leaderboard_cache` (Redis Sorted Sets: `ZADD leaderboard:global:2026-w40 <points> <user_id>`) — Instant sub-5ms ranking queries via `ZREVRANGE`.

6. 8-Week MVP Implementation Roadmap

Sprint / Phase	Duration	Key Deliverables
Sprint 1: Architecture & Data Ingestion	Weeks 1 – 2	• PostgreSQL + PostGIS schema setup & Redis cache layer. • LTA DataMall proxy service (Stops, Routes, Live Arrivals, Alerts). • User Auth (Apple & Google Sign-In) & Cloud CI/CD repository setup.
Sprint 2: Mobile Core & Verification Engine	Weeks 3 – 4	• Flutter UI: Home Dashboard, Route Planner, Live Bus Screen with crowd badges. • Ride Verification Engine implementation (7-Factor Scoring). • Basic Fare-Smart distance fare calculator & transfer warnings.
Sprint 3: Gamification & Admin Portal	Weeks 5 – 6	• Green Points ledger, streaks, badges, and user level progression. • Redis-backed live leaderboards (Global, Weekly, Friends). • Next.js Admin Portal: Contest builder, banner manager, fraud review.
Sprint 4: Hardening & App Store Deployment	Weeks 7 – 8	• Anti-spoofing tests, geofence tuning, load testing. • TestFlight (iOS) and Internal Testing (Google Play) builds. • Production launch handover & complete documentation.

7. Commercial Models & Advance Terms

Option A: Full Turnkey MVP Delivery (Recommended)

Total Project Investment: SGD 15,000 *(or INR 9,50,000)*

Milestone & Advance Payment Structure:

1. **35% Advance upon Project Kickoff: SGD 5,250** *(~INR 3,32,000)* — Paid upfront to initiate architecture lock, CI/CD pipeline setup, and Sprint 1.
2. **30% Milestone 2 Delivery: SGD 4,500** *(~INR 2,85,000)* — Delivered working backend proxy + PostGIS + Flutter MVP with live arrivals & ride verification.
3. **20% Milestone 3 Delivery: SGD 3,000** *(~INR 1,90,000)* — Delivered Gamification engine + Redis leaderboards + Admin Portal.
4. **15% Final Handover: SGD 2,250** *(~INR 1,42,000)* — App Store & Play Store deployment, complete IP & repository handover.

Option B: Monthly CTO & Technical Leadership Retainer

Retainer Fee: SGD 4,000 / month *(~INR 2,50,000 / month)*

Terms: 100% advance at the beginning of each 30-day billing cycle (2-month minimum commitment).

Option C: Core Backend & Verification Engine Only

Total Investment: SGD 9,500 *(40% advance upon signing: SGD 3,800; 40% on API sandbox release; 20% on final handover)*.

Note: Third-party infrastructure costs (Apple Developer Program $99/yr, Google Play $25, AWS/GCP hosting, domain, and official LTA API credentials) are provisioned directly under Aveek's / TransportLah's corporate entity.